AGGREGATE SHOCKS AND DEBT MANAGEMENT I

Fiscal and Monetary Policy 2026

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PLAN

- How should the government change taxes and debt in response to changes in purchases?
- For example: should it finance war spending in the same way as public education?
- This is a normative question, but to the extent real world governments behave according to prescriptions of models we will see in class, it will also be a positive theory of government debt and taxes.

PLAN

- We will study optimal taxation and debt policy in two environments:
  - incomplete markets: the government can buy or sell only a limited set of securities, often only a single risk-free security.
  - complete markets: the government can buy or sell claims contingent on all possible states of the world.
- First study a reduced form model (Barro, 1979), then proceed to optimal taxation in a competitive equilibrium (Lucas and Stokey, 1983).

REDUCED FORM MODELS

INCOMPLETE MARKETS

- We study the following problem based on Barro (1979):
  - the government uses taxes τ_t to finance *stochastic* purchases g_t;
  - $D(\tau_t)$ is the deadweight loss of taxes;
  - the government wants to finance g_t in a way that minimizes the expected value of present discounted deadweight loss of taxes:

$$\min_{\{\tau_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left[\left(\frac{1}{1+r} \right)^t D(\tau_t) \right]$$

  where $1/(1+r)$ is the (gross) rate at which the government discounts the future.

INCOMPLETE MARKETS

- Assume that g_t is governed by an N state Markov chain.
- The state of the world is given by s_t that follows a Markov chain with a transition probability matrix P:

$$P_{ij} = P\left\{s_{t+1} = \bar{s}_j \mid s_t = \bar{s}_i\right\}.$$

- Government spending $\{g_t\}$ obeys

$$g_t = \begin{cases} g_1 & \text{if } s_t = \bar{s}_1 \\ \vdots & \\ g_N & \text{if } s_t = \bar{s}_N. \end{cases}$$

INCOMPLETE MARKETS

- The government can issue only one period, non-state contingent debt (incomplete markets) at the price $1/(1+r)$, equal to the discount rate.
- Government problem:

$$\min_{\{\tau_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left[\left(\frac{1}{1+r} \right)^t D(\tau_t) \right]$$

$$\text{s.t.} \quad g_t + b_t = \frac{1}{1+r} b_{t+1} + \tau_t$$

and, in addition, we assume

$$\mathbb{E}_0 \left[\left(\frac{1}{1+r} \right)^t b_t^2 \right] \leq \infty.$$

RECURSIVE FORMULATION OF THE GOVERNMENT'S PROBLEM

- Write the problem recursively:

$$V(b,g) = \min_{\tau} D(\tau) + \frac{1}{1+r}\mathbb{E}\left[V\left(b',g'\right) \mid g\right]$$
$$\text{s.t.} \quad g + b = \frac{1}{1+r}b' + \tau$$

where x denotes current variables and x' next-period variables.

THE FOC EQUATES MARGINAL DISTORTION TO THE SHADOW VALUE OF DEBT

- Substitute the constraint and differentiate with respect to τ:

$$D'(\tau) = \mathbb{E}\left[\frac{\partial V(b', g')}{\partial b'} \,\middle|\, g\right]$$

- The envelope condition gives us $V'(b, g) = D'(\tau)$, so

$$V'(b', g') = D'(\tau').$$

THE EULER EQUATION: MARGINAL DISTORTION TODAY = EXPECTED MARGINAL DISTORTION TOMORROW

- Combining the FOC with the envelope condition:

$$D'(\tau_t) = \mathbb{E}_t\left[D'(\tau_{t+1})\right]$$

- Tax smoothing: spread out distortionary effects of taxes across time.

INCOMPLETE MARKETS

- There are two special cases we will consider in more detail:
  - Special case 1 - no uncertainty (Barro, 1979):

$$\tau_t = \tau_{t+1}$$

  - Special case 2 - quadratic deadweight loss $(D(\tau_t) = a\tau_t^2 + \gamma)$:

$$\tau_t = \mathbb{E}_t\left[\tau_{t+1}\right]$$

SPECIAL CASE 1: PERFECT FORESIGHT

- With no uncertainty about future g_t (perfect foresight), taxes are constant.
- τ (a single value) is chosen to satisfy the budget constraint:

$$\tau = \frac{r}{1+r}\left[\sum_{t=0}^{\infty}\left(\frac{1}{1+r}\right)^{t} g_t + b_0\right]$$

- Debt absorbs all fluctuations in government spending:

$$b_{t+1} - b_t = r\left(b_t - b_0\right) + \left(1+r\right) g_t - r\sum_{s=0}^{\infty}\left(\frac{1}{1+r}\right)^{s} g_s.$$

SPECIAL CASE 1: PERFECT FORESIGHT

- If $g_t = g$ is constant, then
$$\tau = g + \frac{r}{1+r} b_0, \quad b_t = b_0;$$
debt remains constant at its initial level.
- If $g_0 = g + \epsilon$ and $g_t = g$ for all $t \geq 1$ then
$$\tau = g + \frac{r}{1+r} (\epsilon + b_0), \quad b_t = b_0 + \epsilon \text{ for } t \geq 1;$$
debt increases by ϵ and stays elevated forever.
- The level of debt b_t is irrelevant. What matters for debt is transitory fluctuations in g_t, not the average level of g_t.
- Intuition valid also with trend growth of GDP and government spending (see Barro's paper).

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- What does the condition

$$\tau_t = \mathbb{E}_t\left[\tau_{t+1}\right]$$

  tell us about the optimal tax policy?

- τ_t is a random walk:

$$\begin{aligned}\tau_{t+1} - \tau_t &= \tau_{t+1} - \mathbb{E}_t\left[\tau_{t+1}\right] \\ &= \tau_{t+1} - \left(\tau_{t+1} - \epsilon_{t+1}\right) \\ &= \epsilon_{t+1}\end{aligned}$$

  where ϵ_{t+1} is a martingale-difference innovation.

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- Since $\mathbb{E}_{t-1}[\tau_t] = \tau_{t-1}$, the innovation $\tau_t - \mathbb{E}_{t-1}[\tau_t]$ moves only because of news about expected future g_t.
- In a persistent but non-absorbing high-spending state, observing the state survive one more period is itself new information.
- To see this, use the budget constraint together with $\mathbb{E}_t[\tau_{t+1}] = \tau_t$ to get

$$\tau_t = \frac{r}{1+r} b_t + \frac{r}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r}\right)^s \mathbb{E}_t[g_{t+s}].$$

so

$$\tau_t - \mathbb{E}_{t-1}[\tau_t] = \frac{r}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r}\right)^s \left(\mathbb{E}_t[g_{t+s}] - \mathbb{E}_{t-1}[g_{t+s}]\right)$$

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- The size of the jump in τ_t depends on the size of the change in expected future g_t:
  - If there is a permanent shift in g_t, $\mathbb{E}_t\left[g_{t+s}\right] - \mathbb{E}_{t-1}\left[g_{t+s}\right] = \epsilon$ for all s, taxes change by exactly the same amount ϵ.
  - If the change is temporary, $\mathbb{E}_t\left[g_{t+s}\right] - \mathbb{E}_{t-1}\left[g_{t+s}\right] = \epsilon$ only for $s = 0$, taxes change by $\epsilon r/\left(1+r\right)$.
- Remaining needed resources raised through debt issuance:
  - Debt remains constant for a permanent change in g_t.
  - Debt changes by ϵ for a temporary change in g_t.
- Implication: finance education by taxes, finance war by debt.

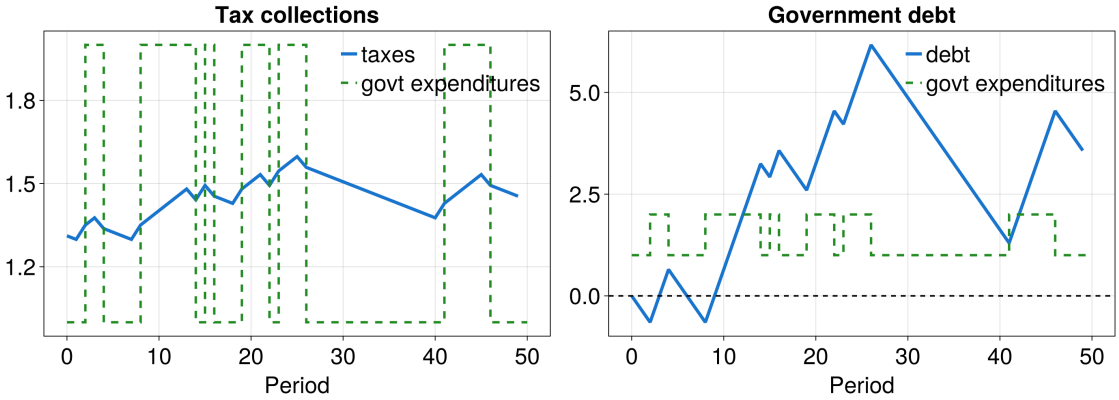
SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- Debt is the residual financing object:

$$b_{t+1} = (1 + r)(g_t + b_t - \tau_t).$$

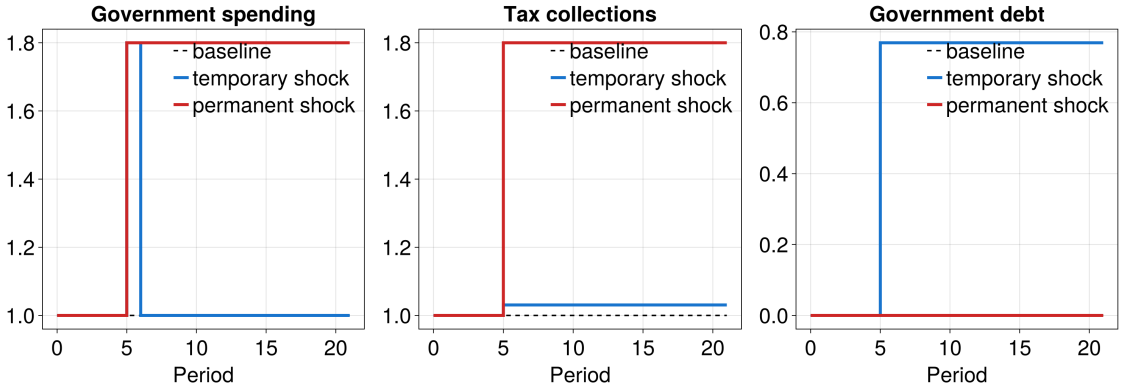
- There is nothing that acts as a force that would push debt to a particular level:
  - No debt target;
  - No "raise taxes if debt is too high" rule;
  - Debt need not revert to any stable level; it can drift for long periods as tax smoothing shifts shocks into future debt service.

NUMERICAL EXAMPLE: BARRO INCOMPLETE MARKETS



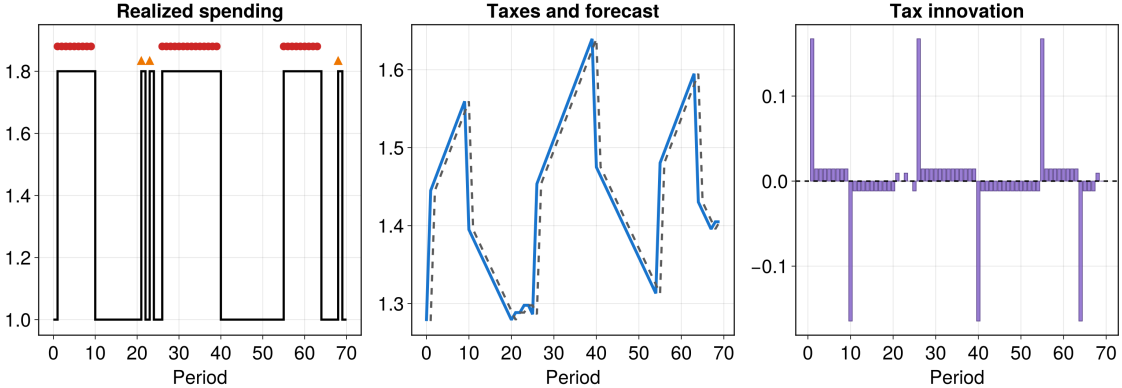
Quadratic tax loss: $D(\tau) = a\tau^2 + \gamma$. Parameters: $\beta = 0.96$, $g_t \in \{1, 2\}$, $P_{12} = 0.2$, $P_{21} = 0.4$, initial debt $b_0 = 0$.

TEMPORARY VS PERMANENT SPENDING SHOCKS



Temporary shock: $\Delta\tau = \epsilon r/(1+r)$ and $\Delta b = \epsilon$. Permanent shock: $\Delta\tau = \epsilon$ and debt is unchanged.

THREE-STATE MARKOV SIMULATION: TAX INNOVATIONS



Dashed gray line: last period's forecast $\mathbb{E}_{t-1}[\tau_t] = \tau_{t-1}$. Purple bars: tax innovation $\tau_t - \mathbb{E}_{t-1}[\tau_t]$. Red circles mark persistent-shock periods; orange triangles mark temporary-shock periods. Both shock states have the same current size, $g_t = \bar{g} + \epsilon$, but state 2 is non-absorbing, so surviving one more period in that state is itself news.

LUCAS AND STOKEY (1983)

SETUP

- We study the following problem based on Lucas and Stokey (1983):
  - the representative household has preferences over consumption and labor (leisure);
  - production technology is linear in labor, there is no capital;
  - the government uses distortionary tax rate τ_t on labor income to finance *stochastic* purchases g_t (we assume g_t is a Markov process);
  - the households and the government can trade a full set of Arrow securities (state-contingent claims).
  - the government wants to finance g_t in a way that maximizes the welfare of the representative household.
- General equilibrium.

SETUP

- Price of securities no longer exogenous.
- The government's cares about welfare of the representative household.
- Think of the problem in two steps:
  1. for given government policies there exists some competitive equilibrium;
  2. the government picks policies that result in the "best" equilibrium.
- Key difference: policies not only have to satisfy the government's budget constraint, but also household's optimality conditions.
- Primal approach: we will look directly for allocations that maximize welfare and then think of policies that implement that.

THE PRIMAL APPROACH IN FOUR STEPS

1. Write down the household problem with Arrow securities;
2. Derive the household first order conditions;
3. Substitute out prices and policies $\Rightarrow$ implementability constraint: a single restriction on allocations that summarizes private optimality;
4. Formulate the Ramsey problem: maximize welfare subject to implementability + resource constraints.

NOTATION: CURRENT STATE VS. HISTORY

- s_t denotes the current exogenous state at date t.
- $s^t \equiv (s_0, \ldots, s_t)$ denotes the entire history up to date t.
- An object indexed by s^t may in principle depend on the whole history:

$$c\left(s^t\right),\ \ell\left(s^t\right),\ \tau\left(s^t\right),\ a\left(s^{t+1}\right),\ q\left(s^{t+1} \mid s^t\right).$$

- The notation $s^{t+1} \geq s^t$ means that s^{t+1} is a successor history that extends the current history by one state realization.
- Since g_t is Markov, probabilities depend on the history only through the last state:

$$\pi\left(s^{t+1} \mid s^t\right) = \pi\left(s_{t+1} \mid s_t\right).$$

- Later we show that under the Ramsey allocation many objects become history independent, so they can be written as functions of s_t alone.

Competitive equilibrium

A competitive equilibrium given government policies $\tau\left(s^t\right)$ and $g\left(s^t\right)$ is a set of allocations $\left(c\left(s^t\right), \ell\left(s^t\right), a\left(s^{t+1}\right)\right)$ and prices $\left(q\left(s^{t+1} \mid s^t\right), w\left(s^t\right)\right)$ such that:

1. Given prices, allocations solve the household problem;
2. Given prices, allocations solve the firm problem;
3. The government budget constraint is satisfied is each state;
4. All markets clear in each state.

- Note: this is not a proper definition of a competitive equilibrium, but it will do for our purposes.

STEP 1: THE HOUSEHOLD PROBLEM WITH ARROW SECURITIES

- Household chooses consumption, labor supply and portfolio of Arrow securities to maximize expected utility:

$$
\begin{aligned}
& \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u\left(c\left(s^t\right), \ell\left(s^t\right)\right) \\
\text{s.t.} \quad \forall s^t \quad & c\left(s^t\right) + \sum_{s^{t+1} \geq s^t} q\left(s^{t+1} \mid s^t\right) a\left(s^{t+1}\right) \\
& = \left(1 - \tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) + a\left(s^t\right) \\
& a\left(s^0\right) = a_0
\end{aligned}
$$

where $u(\cdot)$ is the utility function, $c(\cdot)$ consumption and $\ell(\cdot)$ labor.

STEP 1: COMPLETE MARKETS COLLAPSE TO A SINGLE BUDGET CONSTRAINT

- Impose the natural debt limit: in each state s^t, the household cannot borrow more than the present value of its minimum future after-tax income:

$$a\left(s^t\right) \geq -\bar{A}\left(s^t\right).$$

- This is the tightest borrowing limit that does not shrink the budget set relative to Arrow-Debreu.
- Iterate the sequential budget constraint forward T periods. The natural debt limit ensures that the terminal term $\sum_{s^{T+1}} q^0\left(s^{T+1}\right) a\left(s^{T+1}\right) \geq 0$, where $q^0\left(s^t\right)$ is the time-0 Arrow-Debreu price.
- Passing to the limit, the present-value budget constraint follows:

$$\sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right) c\left(s^t\right) \leq \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) + q^0\left(s^0\right) a_0.$$

STEP 1: THE HOUSEHOLD PROBLEM WITH A SINGLE CONSTRAINT

- The household problem becomes:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) u\left(c\left(s^t\right), \ell\left(s^t\right)\right)$$

$$\text{s.t.} \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right) c\left(s^t\right) = \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right)$$

$$+ q^0\left(s^0\right) a_0$$

- This is nice because there is only one constraint.

STEP 2: HOUSEHOLD FIRST ORDER CONDITIONS

- Write down the Lagrangian and differentiate with respect to $c\left(s^t\right)$ and $\ell\left(s^t\right)$ to get the first order conditions:

$$\left[c\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_c\left(c\left(s^t\right), \ell\left(s^t\right)\right) = \lambda q^0\left(s^t\right)$$
$$\left[\ell\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_\ell\left(c\left(s^t\right), \ell\left(s^t\right)\right) = -\lambda q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)$$

- Note: the Lagrange multiplier λ is the same for all states s^t.
- To simplify notation

$$\left[c\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_c\left(s^t\right) = \lambda q^0\left(s^t\right)$$
$$\left[\ell\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_\ell\left(s^t\right) = -\lambda q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)$$

STEP 3: SUBSTITUTE OUT PRICES AND POLICIES (NECESSITY)

- Necessity: if $\left(c\left(s^t\right), \ell\left(s^t\right)\right)$ is part of a competitive equilibrium, then it satisfies the implementability constraint.
- From the FOCs, replace $q^0\left(s^t\right)$ in the budget constraint:

$$\text{LHS:} \quad \sum_{t,s^t} q^0\left(s^t\right) c\left(s^t\right) = \frac{1}{\lambda} \sum_{t,s^t} \beta^t \pi\left(s^t\right) u_c\left(s^t\right) c\left(s^t\right)$$

$$\text{RHS:} \quad \sum_{t,s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) = -\frac{1}{\lambda} \sum_{t,s^t} \beta^t \pi\left(s^t\right) u_\ell\left(s^t\right) \ell\left(s^t\right)$$

- All λ's and $q^0\left(s^t\right)$'s cancel. The budget constraint becomes the implementability constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right)\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0 \qquad \square$$

STEP 3: THE IMPLEMENTABILITY CONSTRAINT

- The implementability constraint

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right)\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0$$

  summarizes household optimality conditions and the budget constraint.
- Prices, taxes, and debt have been eliminated. The constraint involves only allocations and the initial condition a_0.
- The government can only choose allocations that satisfy this constraint.

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT

- Sufficiency: if an allocation $\left(c\left(s^t\right), \ell\left(s^t\right)\right)$ satisfies the implementability constraint and the resource constraint, then there exist prices, taxes, and debt such that it is a competitive equilibrium.
- **Proof.** Given an allocation, construct the equilibrium:
  1. Set wages: $w\left(s^t\right) = A$ (from the firm problem).
  2. Set taxes: $\left(1 - \tau\left(s^t\right)\right) = -\frac{u_\ell\left(s^t\right)}{u_c\left(s^t\right)} \frac{1}{A}$ (from the intratemporal FOC).
  3. Set Arrow prices: $q\left(s^{t+1} \mid s^t\right) = \beta\, \pi\left(s_{t+1} \mid s_t\right) \frac{u_c\left(s^{t+1}\right)}{u_c\left(s^t\right)}$.

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT (CONT.)

- By construction, the household FOCs are satisfied.
- We still need to find Arrow security holdings $a\left(s^{t+1}\right)$ that satisfy the household budget constraint for each s^t.
- Define

$$x\left(s^t\right) \equiv \sum_{j=0}^{\infty} \sum_{s^{t+j} \geq s^t} \beta^j \pi\left(s^{t+j} \mid s^t\right)\left[u_c\left(s^{t+j}\right) c\left(s^{t+j}\right) + u_\ell\left(s^{t+j}\right) \ell\left(s^{t+j}\right)\right]$$

  and set $a\left(s^t\right) = x\left(s^t\right) / u_c\left(s^t\right)$. Then the sequential budget constraints are satisfied by construction.
- The implementability constraint guarantees $x\left(s^0\right) = u_c\left(s^0\right) a_0$, so the initial condition is satisfied. □
- The implementability constraint + resource constraint fully characterize the set of competitive equilibrium allocations.

FIRM PROBLEM

- Output in this economy is produced by a representative price-taking firm.
- Linear production function that uses labor as an input.
- Firm problem is almost trivial – it chooses labor to maximize profits:

$$\max_{\ell(s^t)} A\ell\left(s^t\right) - w\left(s^t\right)\ell\left(s^t\right).$$

- In a competitive equilibrium we must have

$$w\left(s^t\right) = A.$$

GOVERNMENT BUDGET CONSTRAINT

- The implementability constraint is the household budget constraint (+ optimality conditions)
- The government should also be constrained by its budget constraint.
- We can ignore it and focus directly on the resource constraint of the economy:

$$A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$$

  where $A\ell\left(s^t\right)$ is the linear production technology.
- Why? Walras' law – the government budget constraint is redundant.

STEP 4: THE RAMSEY PROBLEM

- The government chooses allocations $c\left(s^t\right), \ell\left(s^t\right)$ to maximize the household's utility

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) u\left(c\left(s^t\right), \ell\left(s^t\right)\right)$$

subject to the implementability constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) \left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0$$

and resource constraints:

$$\forall s^t \quad A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$$

STEP 4: FULL COMMITMENT

- Full commitment: the government announces its state-contingent plans at time 0 and cannot change them later.
- Everyone else trusts the government and knows it will stick to its plans.

THE PSEUDO-UTILITY FUNCTION W

- Define the pseudo-utility function:

$$W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) \equiv u\left(c\left(s^t\right), \ell\left(s^t\right)\right) + \Phi\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right]$$

  where Φ is the multiplier on the implementability constraint.

- This absorbs the implementability constraint into the objective. The Ramsey problem becomes:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) \quad \text{s.t.} \quad \forall s^t: \; A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$$

- The only constraints left are the resource constraints — one for each state s^t.

FIRST ORDER CONDITIONS VIA W

- For $s^t > s^0$, the FOCs with respect to $c\left(s^t\right)$ and $\ell\left(s^t\right)$ are:

$$W_c\left(s^t\right) = \mu\left(s^t\right)$$
$$W_\ell\left(s^t\right) = -\mu\left(s^t\right)A$$

  where $\mu\left(s^t\right)$ is the multiplier on the resource constraint for state s^t.

- Written out:

$$u_c\left(s^t\right) + \Phi\left[u_{cc}\left(s^t\right)c\left(s^t\right) + u_c\left(s^t\right) + u_{c\ell}\left(s^t\right)\ell\left(s^t\right)\right] = \mu\left(s^t\right)$$
$$u_\ell\left(s^t\right) + \Phi\left[u_{c\ell}\left(s^t\right)c\left(s^t\right) + u_\ell\left(s^t\right) + u_{\ell\ell}\left(s^t\right)\ell\left(s^t\right)\right] = -\mu\left(s^t\right)A$$

Φ IS THE ONLY LINK BETWEEN STATES

- For each state s^t, we have three equations in three unknowns $c\left(s^t\right), \ell\left(s^t\right), \mu\left(s^t\right)$:
  - two FOCs: $W_c\left(s^t\right) = \mu\left(s^t\right), W_\ell\left(s^t\right) = -\mu\left(s^t\right) A$;
  - one resource constraint: $A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$.
- Φ is the only link between different states s^t.
- Once we know Φ, allocations for each s^t are determined separately.
- For two different histories with the same $g\left(s^t\right)$, the solution must be the same!

IMPLICATIONS

- What are the implications for taxes?
- From household first order condition we have

$$\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)=-\frac{u_\ell\left(s^t\right)}{u_c\left(s^t\right)}$$

and linear production technology implies $w\left(s^t\right)=A$ so

$$1-\tau\left(s^t\right)=-\frac{u_\ell\left(c\left(s^t\right), \ell\left(s^t\right)\right)}{u_c\left(c\left(s^t\right), \ell\left(s^t\right)\right)} \frac{1}{A}.$$

- We have solved for $c\left(s^t\right), \ell\left(s^t\right)$, we can get $\tau\left(s^t\right)$!

IMPLICATIONS

- Since for two different states with the same $g\left(s^t\right)$ consumption and labor are the same – taxes must be the same!
- Conclusion: very different from Barro (1979) – no history dependence.
- If government purchases are i.i.d., taxes are i.i.d. as well.
- Contrast with the incomplete markets case: taxes were a random walk.

TAXES ARE NOT FULLY SMOOTHED IN GENERAL

- Unlike Barro (1979), taxes are not generally constant — they depend on the current state.
- When g_t is high, the government may want to increase labor supply to finance it without a large drop in consumption.

EXAMPLE: SEPARABLE PREFERENCES WITH LINEAR DISUTILITY OF LABOR

- Let $u(c, \ell) = \frac{c^{1-\gamma}}{1-\gamma} - \psi\ell$.
- The pseudo-utility function is

$$W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) = \frac{c\left(s^t\right)^{1-\gamma}}{1-\gamma} - \psi\ell\left(s^t\right) + \Phi\left[c\left(s^t\right)^{1-\gamma} - \psi\ell\left(s^t\right)\right]$$

- The FOCs $W_c = \mu\left(s^t\right)$, $W_\ell = -\mu\left(s^t\right)A$ become:

$$\left[1 + \Phi(1-\gamma)\right] c\left(s^t\right)^{-\gamma} = \mu\left(s^t\right)$$
$$\left[1 + \Phi\right]\psi = \mu\left(s^t\right)A$$

EXAMPLE

- For each state $s^t > s^0$ we have:

$$\frac{1+\Phi}{[1+\Phi(1-\gamma)]A}\psi = c\left(s^t\right)^{-\gamma}$$

  so consumption is constant.

- Taxes satisfy

$$1-\tau\left(s^t\right) = \frac{\psi}{c\left(s^t\right)^{-\gamma}}\frac{1}{A}:$$

  since $c\left(s^t\right)$ is constant, taxes are constant as well.

- With these preferences households supply labor fully elastically, fluctuations in g_t are absorbed by changes in production.

LESSONS

- Remember: the above result is not general.
- What is general is that allocations and taxes are history independent — they are fully determined by the current state s_t.
- What about the optimal path of debt?

ARROW PRICES ARE PINNED DOWN BY ALLOCATIONS

- From the household FOC for Arrow securities:

$$q\left(s^{t+1} \mid s^{t}\right) = \beta\, \pi\left(s_{t+1} \mid s_{t}\right) \frac{u_c\left(s^{t+1}\right)}{u_c\left(s^{t}\right)}$$

- Since allocations depend only on s_t, prices depend only on (s_t, s_{t+1}).
- Once we solve the Ramsey problem for allocations, all prices are determined.

TIMING WITHIN DATE t

- Current history s^t is observed.
- The inherited payoff term $b\left(s^t\right)$ from the portfolio chosen at $t-1$ comes due at date t.
- Taxes are collected and government spending $g\left(s^t\right)$ is paid.
- The government then chooses the next-period Arrow portfolio $\left\{b\left(s^{t+1}\right)\right\}_{s_{t+1}}$ and receives its date-t sale value $\sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$.

THE GOVERNMENT BUDGET CONSTRAINT PINS DOWN PORTFOLIO VALUE

- The government budget constraint for each history s^t:

$$g\left(s^t\right) + b\left(s^t\right) = \tau\left(s^t\right) A \ell\left(s^t\right) + \sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$$

  where $b\left(s^t\right)$ is the payoff due today from the portfolio chosen at $t-1$, and $\left\{b\left(s^{t+1}\right)\right\}_{s_{t+1}}$ is the payoff menu chosen today for $t+1$.
- We know $g\left(s^t\right), \tau\left(s^t\right), \ell\left(s^t\right)$, and $q\left(s^{t+1} \mid s^t\right)$ from the Ramsey solution.
- The budget constraint pins down the date-t market value of the continuation portfolio, $\sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$.
- The full state-by-state payoff menu is recovered on the next slides using the recursion for $x\left(s^t\right)$.

DEFINE MARGINAL-UTILITY-ADJUSTED DEBT $x\left(s^t\right) \equiv u_c\left(s^t\right) b\left(s^t\right)$

- Multiply the government budget constraint by $u_c\left(s^t\right)$ and substitute Arrow prices:

$$x\left(s^t\right) = u_c\left(s^t\right)\left[\tau\left(s^t\right) A\,\ell\left(s^t\right) - g\left(s^t\right)\right] + \beta \sum_{s_{t+1}} \pi\left(s_{t+1} \mid s_t\right) x\left(s^{t+1}\right)$$

- Since allocations depend only on s_t, guess that $x\left(s^t\right)$ depends only on s_t. Then in vector form:

$$\vec{x} = \vec{h} + \beta\,\Pi\,\vec{x}$$

where $h\left(s\right) \equiv u_c\left(s\right)\left[\tau\left(s\right) A\,\ell\left(s\right) - g\left(s\right)\right]$ and Π is the Markov transition matrix.

RECOVERING THE NEXT-PERIOD PAYOFF MENU

- Solve the linear system:

$$\vec{x} = (I - \beta \Pi)^{-1} \vec{h}$$

- Recover the payoff promised for any realized state s:

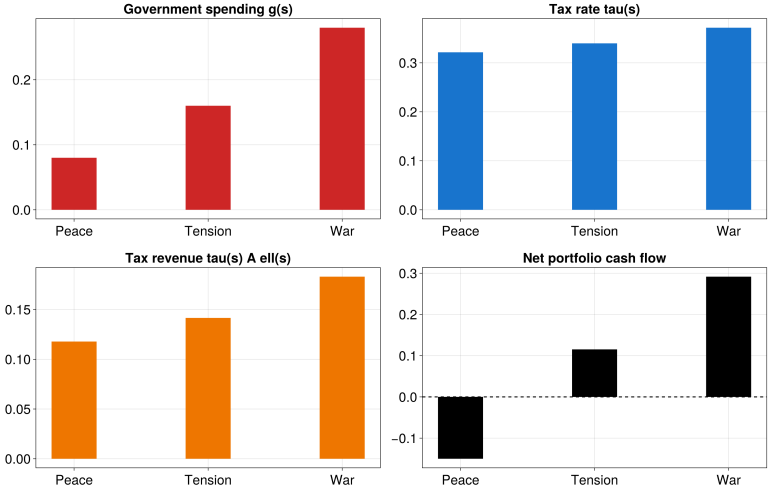
$$b(s) = \frac{x(s)}{u_c(s)}$$

- After history s^t, the inherited payoff due today is $b(s_t)$.
- At date t, the government then issues the whole menu $\{b(s_{t+1})\}_{s_{t+1}}$ for date $t+1$.
- Each promised payoff depends only on the successor state s_{t+1}, not on the full history s^t: the issued menu is history independent.

WHAT THE DEBT OBJECTS MEAN

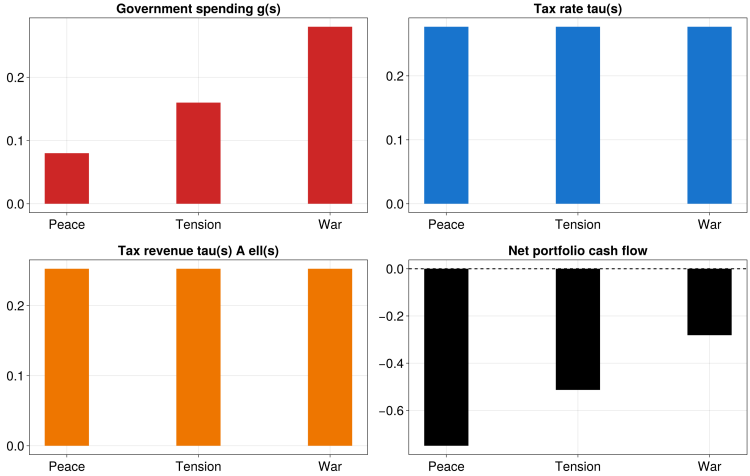
- Inherited payoff due today after history s^t: $b(s_t)$.
- New payoff menu chosen today for tomorrow: $\{b(s_{t+1})\}_{s_{t+1}}$.
- $V(s_t) \equiv \sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b(s_{t+1})$: the market value today of that new continuation portfolio.
- $x(s_t) \equiv u_c(s_t)\, b(s_t)$: an auxiliary object that makes the recursion linear. It is useful for solving the model, but less important economically than the payoff menu itself.
- For intuition, it is often clearer to plot net portfolio cash flow to the government, $-b(s_t)$: positive means the portfolio pays the government, negative means the government makes a payment.

A FINITE-STATE LUCAS-STOKEY EXAMPLE



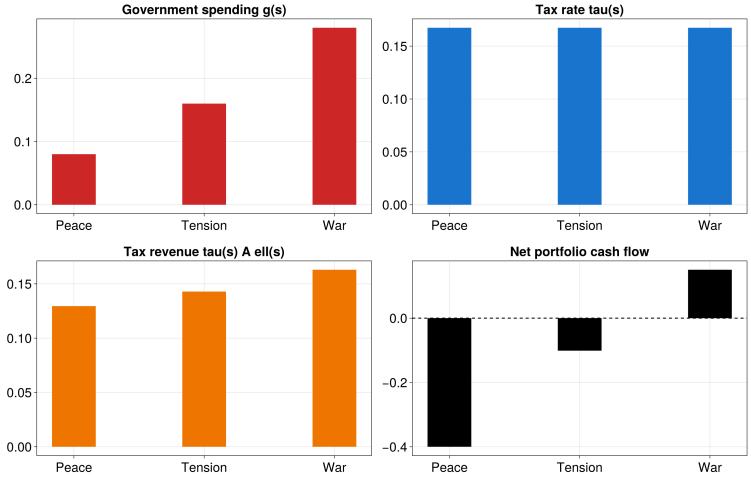
Three-state Markov example with $u(c, \ell) = \log c + \psi \log(1 - \ell)$ and target peace-state payoff $b(\text{peace}) = 0.15$. The four panels show government spending, the tax rate, tax revenue, and net portfolio cash flow $-b(s)$, where positive values mean receipts by the government.

ALTERNATIVE EXAMPLE: QUASI-LINEAR UTILITY



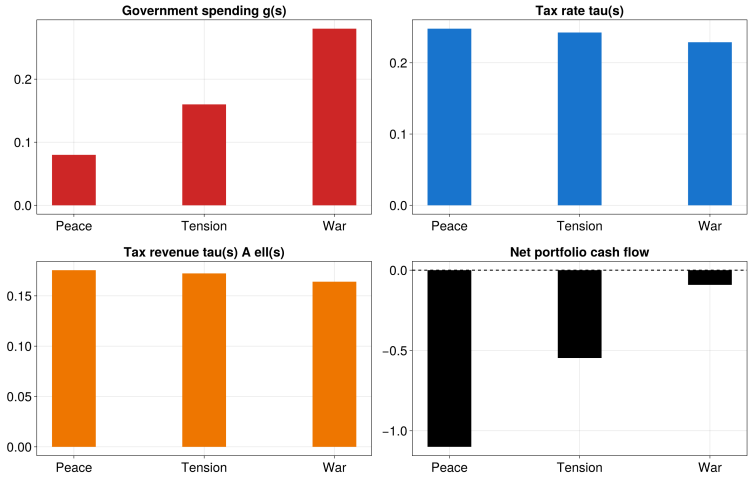
With $u(c, \ell) = c - \psi \ell^{1+\chi}/(1+\chi)$ and high labor curvature $\chi = 8$, u_c is constant. Tax rates are flat across states, while net portfolio cash flow adjusts to absorb the fiscal shock.

ALTERNATIVE EXAMPLE: LOG UTILITY, LINEAR LABOR



With $u(c, \ell) = \log c - \psi\ell$, the tax rate is constant across states, while tax revenue and net portfolio cash flow respond to g.

ALTERNATIVE EXAMPLE: GHH UTILITY

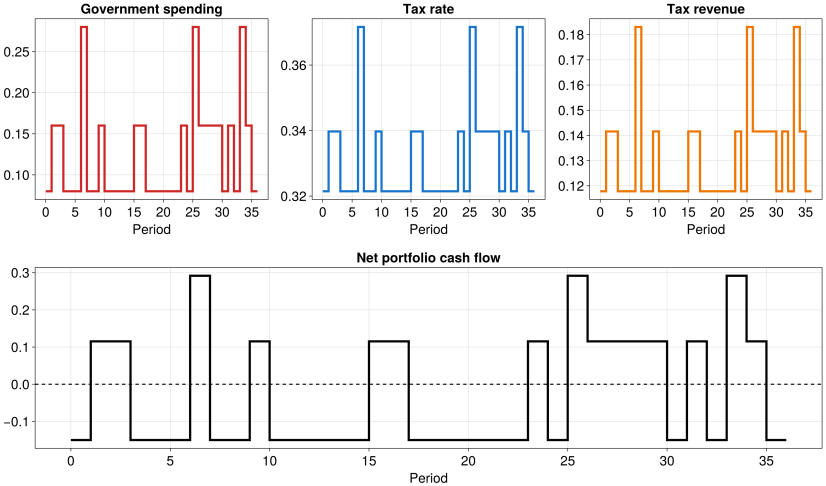


With $u(c, \ell) = \log\big(c - \kappa \ell^{1+\nu}/(1+\nu)\big)$, the tax formula becomes $1 - \tau(s) = \kappa \ell(s)^{\nu}/A$. In this calibration, high g lowers tax rates and raises net portfolio receipts.

STATE-CONTINGENT DEBT AS INSURANCE

- The government chooses a portfolio of state-contingent positions — a separate payoff for each possible next-period state s_{t+1}.
- In the examples, the net portfolio cash flow is more positive in high-g states and more negative in low-g states.
- Effect: the government hedges fiscal shocks, helping keep taxes smoother across states.
- Contrast with incomplete markets: without state-contingent debt, the government cannot hedge $\Rightarrow$ taxes drift as a random walk.

HISTORY INDEPENDENCE IN A SIMULATED PATH



A simulated path from the same three-state Markov chain. Whenever the economy revisits a state, the same spending realization, tax rate, tax revenue, and net portfolio cash flow recur.

TIME INCONSISTENCY AND PORTFOLIO COMMITMENT

THE TIME-0 FOC IS DIFFERENT

- The FOC for $c\left(s^0\right)$ at $t = 0$ has an extra term compared to $t \geq 1$:

$$W_c\left(s^0\right) \underbrace{-\Phi\, u_{cc}\left(s^0\right) a_0}_{\text{extra term}} = \mu\left(s^0\right)$$

$$W_\ell\left(s^0\right) \underbrace{-\Phi\, u_{c\ell}\left(s^0\right) a_0}_{\text{extra term}} = -\mu\left(s^0\right) A$$

- This term comes from differentiating the $u_c\left(s^0\right) a_0$ term in the implementability constraint.
- Time-0 allocations differ from $t \geq 1$, even with the same g.

THE CAPITAL LEVY ON INITIAL DEBT

- The extra term gives the time-0 government an incentive to manipulate $u_c\left(s^0\right)$ to reduce the real value of inherited debt a_0.
  - For example: policies that raise $c\left(s^0\right)$ lower $u_c\left(s^0\right)$, making debt cheaper to repay.
  - This is a capital levy: expropriating initial debt holders through the price level or tax policy.
- This is good for welfare: it allows lower tax distortions in all future periods.
- The continuation planner ($t \geq 1$) takes $x_t = u_c\left(s^t\right)\, b\left(s^t\right)$ as given and cannot manipulate it. The capital levy is available only at $t = 0$.

WHY THE RAMSEY PLAN IS TIME-INCONSISTENT

- Suppose the government could re-optimize at $t = 1$.
- It would face the same temptation: inherited debt $b\left(s^1\right)$ is measured in goods, and the government can manipulate $u_c\left(s^1\right)$ to reduce its burden.
- Period 1 would "look like period 0" — the same extra term appears.
- A continuation of the time-0 Ramsey plan is not a Ramsey plan for the continuation problem.
- Reason: the original plan takes $x_1\left(s_1\right) = u_c\left(s^1\right)\ b\left(s^1\right)$ as given, but a re-optimizing government takes $b\left(s^1\right)$ as given and can alter $u_c\left(s^1\right)$.

WHEN IS THE RAMSEY PLAN TIME-CONSISTENT?

- The extra term $-\Phi\, u_{cc}\left(s^0\right)\, a_0$ vanishes in three special cases:
  1. $a_0 = 0$: no initial debt to levy on.
  2. u_c is constant (quasi-linear preferences): the government cannot manipulate marginal utility.
  3. $\Phi = 0$: initial government assets are large enough that the first-best is attainable.
- Outside these cases, the Ramsey plan is time-inconsistent.
- This is the central problem of Lucas and Stokey (1983).

CONTINUATION PLANNER AFTER A HISTORY s^t

- Write the continuation problem after a realized history $s^t = (s_0, \ldots, s_t)$.
- The inherited endogenous state is $x\left(s^t\right) \equiv u_c\left(s^t\right) b\left(s^t\right)$.
- The planner chooses current labor $\ell\left(s^t\right)$ and a full menu of promised continuation states $\left\{x\left(s^{t+1}\right)\right\}_{s^{t+1} \geq s^t}$.

$$V\left(x\left(s^t\right), s^t\right) = \max_{\ell(s^t),\, \{x(s^{t+1})\}_{s^{t+1} \geq s^t}} u\left(A\ell\left(s^t\right) - g\left(s^t\right), \ell\left(s^t\right)\right) + \beta \sum_{s_{t+1}} \pi\left(s_{t+1} \mid s_t\right) V\left(x\left(s^{t+1}\right), s^{t+1}\right)$$

subject to

$$x\left(s^t\right) = u_c\left(s^t\right)\left(A\ell\left(s^t\right) - g\left(s^t\right)\right) + u_\ell\left(s^t\right) \ell\left(s^t\right) + \beta \sum_{s_{t+1}} \pi\left(s_{t+1} \mid s_t\right) x\left(s^{t+1}\right).$$

- Because g_t is Markov, future probabilities depend on the history only through the last state s_t, even though we keep the full history in the notation.

TIME-0 PLANNER: INHERITED DEBT IS b_0, NOT $x\left(s^0\right)$

- At time 0, the government inherits goods debt b_0.
- It does not inherit $x\left(s^0\right)$, because $x\left(s^0\right) = u_c\left(s^0\right) b_0$ depends on the current allocation choice.

$$W\left(b_0, s^0\right) = \max_{\ell\left(s^0\right), \{x\left(s^1\right)\}_{s^1 \geq s^0}} u\left(A\ell\left(s^0\right) - g\left(s^0\right), \ell\left(s^0\right)\right) + \beta \sum_{s_1} \pi\left(s_1 \mid s_0\right) V\left(x\left(s^1\right), s^1\right)$$

subject to

$$u_c\left(s^0\right) b_0 = u_c\left(s^0\right)\left(A\ell\left(s^0\right) - g\left(s^0\right)\right) + u_\ell\left(s^0\right) \ell\left(s^0\right) + \beta \sum_{s_1} \pi\left(s_1 \mid s_0\right) x\left(s^1\right).$$

- This is why period 0 is special: the planner can manipulate $u_c\left(s^0\right)$ and therefore the inherited burden $u_c\left(s^0\right) b_0$.

CHOOSING PROMISED CONTINUATION STATES HISTORY BY HISTORY

- Fix a current history s^t. The continuation planner chooses one promise $x\left(s^{t+1}\right)$ for each successor history $s^{t+1} \geq s^t$.
- The first order condition with respect to each promise is

$$V_x\left(x\left(s^{t+1}\right), s^{t+1}\right) = \Phi\left(s^t\right) \qquad \forall s^{t+1} \geq s^t,$$

  where $\Phi\left(s^t\right)$ is the multiplier on the recursive implementability constraint at history s^t.
- So all successor histories branching from the same current history share the same shadow value $\Phi\left(s^t\right)$.
- Only after writing the problem this way can we use the Markov structure to conclude that, in equilibrium, $x\left(s^t\right)$ can be written as a function of the current state s_t alone.

DEBT PORTFOLIO AS A COMMITMENT DEVICE

- At time 0, the government chooses a full menu of one-period Arrow claims $\left\{b\left(s^1\right)\right\}_{s^1 \geq s^0}$, one for each successor history s^1.
- Equivalently, it chooses the inherited continuation states

$$x\left(s^1\right) \equiv u_c\left(s^1\right) b\left(s^1\right) \qquad \text{for all } s^1 \geq s^0.$$

- After one particular history s^1 is realized, the continuation planner at that history takes that inherited value $x\left(s^1\right)$ as given.
- It can choose future promises $\left\{x\left(s^2\right)\right\}_{s^2 \geq s^1}$, but it cannot go back and change the inherited object $x\left(s^1\right)$.

DEBT PORTFOLIO AS A COMMITMENT DEVICE

- If the time-0 planner picks $\left\{x\left(s^1\right)\right\}_{s^1 \geq s^0}$ so that

$$V_x\left(x\left(s^1\right), s^1\right) = \Phi\left(s^0\right) \qquad \forall s^1 \geq s^0,$$

  then every continuation planner voluntarily implements the promised Ramsey continuation.
- This is the commitment role of the debt portfolio: the menu chosen at time 0 pins down the inherited state variable at each successor history.
- The continuation planner is free to optimize going forward, but not to undo the inherited portfolio position chosen earlier.

SUMMARY: THE LUCAS-STOKEY RESULT

- The Ramsey plan is generically time-inconsistent: the government wants to levy a capital tax on initial debt holders.
- But with complete markets, the government can choose a state-contingent debt portfolio that removes the temptation to re-optimize.
- The key variable is $x(s) = u_c(s)\ b(s)$: marginal-utility-adjusted debt.
- The time-0 planner sets $x(s)$ for each state so that the continuation planner's optimum reproduces the original plan.
- Without complete markets (Aiyagari, Marcet, Sargent, Seppälä, 2002), the multiplier Φ becomes time-varying and history-dependent — the government cannot fully commit via the debt portfolio.